Data Mining Group Project Report
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My analysis was to try to discover which products are frequently purchased together. This is useful for amazon because they can then suggest products and remind customers that they would want those to, furthermore it enables targeted discount or bundles. The Association Rules model purpose is to identify associations between specific items by analyzing how frequently they are purchased together. The model creates a matrix with a set of every customer and each product with just one score. Either has the item or does not have the item. Then Apriori scans your basket matrix and finds item sets (products) that appear together often enough. For this project I conducted three rounds of association. Round 1; I used the orderID and productID searching for frequently associated products within individual orders, the results revealed a lack of meaningful or interesting patterns. For round 2; I decided to assess customer ID to Product, thinking that perhaps customers that order only one thing at a time still have certain products associated across their orders overall. When association rule analysis was applied, no association rules were found. Ok, I was almost giving up on finding out, when I decided there is still a business use case for associating categories at a high level. Round 3: Customer ID to category, in this analysis one rule was found. This higher-order association rule shows that customers who purchased Books, Home & Kitchen, Clothing, and Toys & Games together were more likely to also purchase Electronics. (confidence = 0.33, lift = 1.01). This means that 33% of such customers also bought Electronics, but the lift value close to 1 indicates that this relationship is very weak and close to random. These results suggest that there is a high overall frequency of categories in customer purchases, because people use amazon to buy everything. Although using association rules for marketing and recommendations is a great business tool, this data did not bring any new insights except that customers are purchasing different items and categories at random. Perhaps amazon should recommend items that don't have to do with the product because according to this data it is just as likely for them to purchase an item that is similar to one that is not.

Meira- Random Forest Regression
The model learns how different sales features (price, quantity, category, customer type, etc.) affect the TotalAmount customers spend. These settings let it learn enough detail without becoming too slow or too complex.
RMSE ≈ 21 On average, the model predicts TotalAmount with an error of about 21. If an order is really 350, the model might predict around 329 or 371.
MAE ≈ 15 The average mistake is $15—this is the typical prediction error.
R^2 = 0.9992 This means the model explains 99.92% of the patterns in your Amazon sales. That is extremely good. The model understands almost everything about how the features relate to the final amount the customer pays.
Cross-Validation R^2 = 0.9991 The model is consistent, meaning it performs almost the same on different parts of the dataset

Limitations:
This is extremely high, which is good, but sometimes models that are "too perfect" may be overfitting. It memorizes the data too closely and might not work as well on totally new Amazon data.

Tehila
K means clustering

I used K-Means clustering to see if the data could be grouped in a meaningful way. At first, I ran into a few issues—mainly that the model took about five minutes to run every time I made a change. To speed things up, I switched to Mini-Batch K-Means, which samples 500 data points at a time.

I then experimented with different values of K to find the best number of clusters. The first result suggested just two clusters, but that didn't feel very useful. Increasing K made things a bit blurry—K = 3 gave a clear separation, and K = 4 still showed obvious groups, though slightly less distinct.

With some help from ChatGPT, I added a few new columns to the dataset based on existing data to see if it would reveal new clusters. This helped a bit. Looking at the silhouette and elbow plots together, the silhouette suggested K between 4 and 6, while the elbow plot leaned toward 2. Considering both, I settled on K = 4, which split the data nicely into four clear groups.

Group 0: Premium Shoppers - buy single expensive items, no discounts and low shipping impact.

Group 1: Budget Shoppers- buy low cost items with small orders. Shipping makes up a bigger part of their purchase.

Group 2: Bulk Shoppers-make large orders of semi-expensive items, they are the highest spenders.

Group 3: Mid Range- buy regularly priced things, in regular quantities.

Name	purpose/Q	Success?	Conclusion / business insight
Random Forest Regression	What effects total order amount	yes	Can predict that ___ will cause higher / lower order amount
K Means Clustering	Find meaningful groups	Yes	Divided into 4 groups

Association Rules	What products are associated together	No	Amazon data is so random and vast no associations